

A Lightweight Algorithm for Efficient Synchronization in LoRaWAN Class-B Networks

Luca Scalambri¹, Graduate Student Member, IEEE, Andrea Zanella, Senior Member, IEEE, and Xavier Vilajosana², Senior Member, IEEE

Abstract—The Internet of Things (IoT) has become increasingly relevant in the context of large-scale asset monitoring, revolutionizing the way data is collected and analyzed. In addition to low power consumption and long-range coverage, these applications require precise time synchronization to generate accurate data to assess the structural health of critical assets. Meeting these synchronization requirements is challenging because the devices are battery-powered, deployed in extensive environments, and limited in computational capabilities. LoRaWAN, a widely adopted standard for long-range wireless networks, addresses some of these challenges. In fact, its Class-B operational mode introduces a time synchronization mechanism based on a beacon broadcasting system. However, the energy-intensive nature of this configuration can be inefficient when devices rely on limited power sources. To address this, we introduce a predictive mechanism to dynamically adjust the beacon period in LoRaWAN Class-B networks. Extensive experimentation proves that our approach reduces the average interbeacon synchronization error by three orders of magnitude compared to the LoRaWAN standard. This improvement in timing accuracy enables more efficient energy usage, extending device battery life by more than three times relative to standard configurations. Notably, these gains are achieved with a lightweight implementation requiring only 2.5 KB of memory. Overall, our solution enhances both synchronization performance and energy efficiency in an established standard, making it a practical and cost-effective solution for IoT-based monitoring systems.

Index Terms—Energy efficiency, Internet of Things (IoT), large-scale asset monitoring, LoRaWAN class-B, synchronized wireless networks.

I. INTRODUCTION

IN RECENT years, the Internet of Things (IoT) has gained significant traction [1], primarily due to its ability to facilitate seamless connectivity among devices in complex systems.

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Luca Scalambri is with the Wireless Networks Group, Open University of Catalonia, 08018 Barcelona, Spain, and also with the Department of Engineering, Worldsensing SL, 08014 Barcelona, Spain (e-mail: lscalambri@uoc.edu).

Andrea Zanella is with the Department of Information Engineering, University of Padua, 35131 Padua, Italy (e-mail: andrea.zanella@unipd.it).

Xavier Vilajosana is with the Wireless Networks Group, Open University of Catalonia, 08018 Barcelona, Spain (e-mail: xvilajosana@uoc.edu).

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Another attractive aspect of the IoT paradigm is its versatility, as this technology finds application in a variety of scenarios, including smart cities [2], health care [3], and the monitoring of industrial, civil, and agricultural assets [4], [5], [6].

Specifically, IoT has transformed the field of structural health monitoring (SHM), which focuses on continuously monitoring the structural integrity of infrastructures, by replacing traditional manual processes with fully automated systems. Currently, diverse wireless technologies, including NB-IoT, ZigBee, Wi-Fi, and BLE, are widely employed for standard monitoring applications [7], [8], [9], [10], [11], [12], [13], [14]. However, SHM for critical large-scale assets, including railways, subways, viaducts, bridges, and other similar structures, presents unique and complex challenges that extend beyond the capabilities of conventional IoT systems. These scenarios require energy-efficient devices to minimize deployment costs and long-range communication to reduce device counts [15]. Hence, given that such applications often require data rates that are not exceedingly high, Low Power Wide Area Network (LPWAN) technologies are particularly suitable, given their low power and long-range characteristics, with LoRa/LoRaWAN emerging as one of the most promising ones [16], [17], [18], [19].

In addition to high energy efficiency and long-range coverage, SHM also requires precise time synchronization among devices, essential to generate data that can then be accurately correlated and processed. In fact, most of the monitoring systems used in SHM are configured to periodically sense physical properties, such as deformation, structural strength, or frequency of oscillation. As a result, these networks produce time-series of measurements that are analyzed by inspectors and civil engineers to assess the mechanical performance of the infrastructure [8]. Therefore, time synchronization among the data-collecting sensors is crucial to properly interpret the temporal behavior of the monitored processes and identify potential anomalies.

Unfortunately, time synchronization is difficult to achieve in wireless sensor networks (WSN) because low-cost IoT devices commonly rely on inexpensive electronic components that are susceptible to external factors, such as temperature and humidity, affecting their ability to maintain a stable internal clock. Additionally, absolute time-synchronization solutions based on Internet access or the global positioning system (GPS) are often unavailable on low-cost devices or are too energy-intensive. To address this, LoRaWAN offers an operational mode, known as Class B, which improves device synchronization through a

beacon-broadcasting mechanism [20]. Although this makes it well-suited for applications, such as SHM, it does come with the limitation of consuming more energy compared to the more commonly used Class-A mode.

A. Contributions and Manuscript Outline

In this article, we propose a lightweight mechanism to improve the time synchronization capabilities of LoRaWAN Class-B devices, while reducing the energy consumption. The method dynamically adjusts the beacon period by predicting clock drift between devices based on temperature variations, making it a suitable solution for most SHM applications that require both data synchronization and energy efficiency.

The main contributions of this work can thus be summarized as follows.

- 1) We introduce the first model for Class-B LoRaWAN that captures the communication overhead compared to the traditional Class-A mode, while also quantifying the battery lifetime improvements achieved by extending the system's beacon period.
- 2) We develop a novel model that accurately captures the limitations of real hardware oscillators in constrained devices, specifically focusing on how timekeeping is performed using onboard registers. Unlike existing models in the literature, our approach is closely aligned with the actual behavior of low-power hardware, enabling a more accurate characterization of the synchronization error between devices.
- 3) We propose an adaptive mechanism that dynamically compensates the clock drift and adapts the beacon period of devices, enhancing energy efficiency in Class-B LoRa-based networks deployed in harsh environments.
- 4) We implemented the method as a lightweight algorithm, requiring only 2.5 kB of memory, making it suitable for extremely constrained devices.
- 5) We validated the mechanism in a real-world setting, demonstrating the method's applicability and practicality. Experimental results reveal a reduction of more than three orders of magnitude in the average interpacket synchronization error and more than a three-fold increase in the battery lifetime of the device compared to the LoRaWAN standard, all achieved without requiring any hardware modifications and remaining fully compliant with LoRaWAN specifications.

The remainder of this article is organized as follows. Section II reviews the related work. LoRaWAN is introduced in Section III, with particular emphasis on Class-B mode and its energy-related characteristics. In Section IV, we present a mathematical formulation of the prediction problem, which is addressed by the adaptive solution proposed in Section V. The experimental setup is described in Section VI, and the corresponding results are analyzed in Section VII. Finally, conclusions are discussed in Section VIII.

II. RELATED WORK

With the increasing demand for precise data measurements for remote monitoring of civil and industrial assets, the research community has invested time and effort to develop

different methodologies to counteract the problem of time synchronization in monitoring systems. AbdelRaheem et al. [21] introduced an IoT system for SHM where a set of leaf nodes measure structural parameters and send measurements via Wi-Fi to a collector node, which in turn is connected to a monitoring server via the Internet. Both types of devices, collector and leaf nodes, are equipped with GPS to keep time synchronization among the measurements. This system might be useful in scenarios where leaf nodes are not constrained by energy limitations, but the system becomes impractical with a battery-powered device, as the energy drained by GPS is substantial.

Romanov et al. [22] proposed an innovative low-cost hybrid cross-technology module for precisely synchronized communications. The module consists of two main parts, a system-on-a-chip (SoC) and a low-cost field programmable gate array (FPGA). Although this hybrid system can provide highly accurate synchronization, the energy efficiency is still too low for battery-based devices. In [23], a novel prototype of NB-IoT based on micro-electromechanical systems (MEMS) is proposed to collect vibration-based dynamic measurements, thus improving structural monitoring. However, this solution assumes that all monitored assets are situated in areas where cellular infrastructure is established and operational, while this might not be the case for large-scale facilities that are located outside urban areas, or underground. A cross-technology approach has been applied in [24], [25], and [26] to increase synchronization accuracy of IoT devices. In [24], for example, Gao et al. presented a system denominated *TimeBee*, where a collector node connects to leaf nodes using Wi-Fi. In parallel, similar to the architecture suggested in [27], leaf nodes communicate among themselves through ZigBee, allowing for a more flexible topology. However, this flexibility causes a reduction of the synchronization accuracy, in particular for large multihop networks. Additionally, as Wi-Fi links lack the capability to establish long-range connections in hostile environments, the system becomes unfeasible for synchronized monitoring of large-scale infrastructures.

In their study [28], Fan et al. explored a machine learning strategy involving an enhanced Support Vector Machine technique, which builds upon previous work in [29], to estimate clock offset and drift. This method relies on a large dataset of timestamps gathered between node pairs. However, as typical with data-driven methods, achieving greater accuracy requires a significant volume of data. Consequently, deploying this approach on low-power edge devices used for monitoring turns impractical due to insufficient computational resources and energy availability.

On the other hand, many studies have also focused on the context of LPWANs, specifically targeting data synchronization accuracy. In [30], the authors have proposed a method based on a lightweight algorithm to synchronize data measurements generated by IoT devices in a per hop manner. For this, instead of correcting the nodes' clock, the measurement timestamp created at a certain point in a multihop network is progressively corrected at each hop along the path to the sink node. Despite the fact that this mechanism reduces the typical overhead introduced by synchronization protocols, it is not suitable for low-power networks, as the developed system

assumes that devices continuously listen to the medium to detect new incoming packets, which makes it unfeasible for devices running on batteries.

Rather than focusing on data-level synchronization, several works have explored device-level synchronization with an explicit emphasis on energy efficiency [31], [32], [33], [34]. These approaches are particularly relevant for battery-powered devices with stringent resource constraints, where minimizing energy consumption is critical. To meet both synchronization accuracy and energy-saving goals, techniques, such as PID controllers, temperature-drift lookup tables, and linear regression models are employed to estimate and compensate for clock drift due to temperature variations. By reducing the frequency of synchronization messages, these methods help conserve energy and extend device lifetime. However, they often rely on extensive precalibration and detailed component-level characterization, which limits their scalability and general applicability. Additionally, since many of these methods are not designed for long-range communication, they typically require multihop network architectures to extend coverage. This not only increases system complexity but can also degrade synchronization accuracy and prove impractical for applications, such as SHM.

It is notable that, although the works mentioned above present solid ideas and concepts regarding the use of IoT for synchronized monitoring, to the best of our knowledge, there is no solution for monitoring industrial large-scale infrastructures, able to guarantee robust, reliable and synchronized measurements with low-power and long-range communication technologies. LoRaWAN stands out as a potential exception, as it is a widely deployed industrial standard providing a specific operational mode (Class-B) to increase network synchronization. However, Class-B mode is significantly less energy-efficient than the typical Class-A. In the following section, together with a brief LoRaWAN overview, we present an energy consumption model for both operational modes to illustrate this disparity and introduce an approach that preserves Class-B's synchronization benefits while improving overall energy efficiency.

III. LORAWAN BACKGROUND

LoRa is a proprietary radio modulation engineered to offer Long-Range and Low-Power communications through the use of Chirp Spread Spectrum (CSS) techniques. LoRaWAN, on the other hand, is a media access control (MAC) layer protocol built on top of the LoRa PHY layer [20]. Deployed in star-like topologies, LoRaWAN networks consist of three primary components: 1) End devices; 2) gateways; and 3) a network server. The *network server* serves as the central entity responsible for managing the entire network, while *gateways* are in charge of forwarding packets between the End devices and the network server. *End devices*, on the other hand, are tiny battery-powered devices in charge of measuring data and sending it to the network server through LoRa packets. End devices can operate in three different modes: 1) Class-A; 2) Class-B; and 3) Class-C.

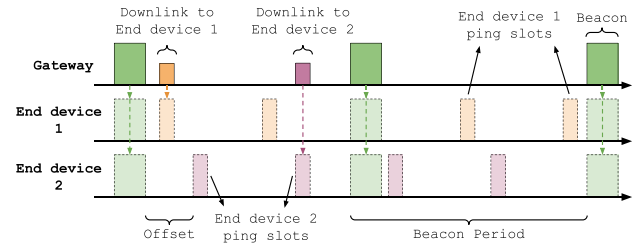


Fig. 1. LoRaWAN network with two Class-B End devices, both having 2 ping slots, but with different offsets.

Class-A devices remain in sleep mode all the time and wake up only when data must be transmitted. Consequently, the network server can only communicate with them after uplink transmission. The last class, Class-C, consists of devices that continuously listen, except during transmission, making them available to the network server at almost any time. However, this modality consumes considerable energy and requires connection to the power grid. Finally, Class-B mode balances energy consumption and reactivity. Since this work focuses in Class-B devices, this modality is better explained in the following.

A. Class-B for Synchronized Networks

Class-B devices behave as Class-A, but they also open additional reception windows to receive periodic beacons and downlink data from the gateways. The standard setting for the beacon period T_{BCN} is 128 s and, within this period, a 5.12 s interval is reserved for beacon reception. This reserved interval ensures that beacon transmissions do not interfere with downlinks that the gateway may send in the assigned slots, effectively eliminating the risk of collisions even under high network loads. The remaining time interval of 122.88 s is divided into $2^{12} = 4096$ slots of 30 ms, which will be grouped into 2^k windows of $W_k = (2^{12}/2^k)$ slots each. The parameter k represents a periodicity factor and can take integer values from 0 to 7, so the number of windows can range from 1 to 128 [35].

The first slot in each window of W_k slots is elected as *ping slot* and used by a Class-B device to receive downlink data from the gateways. To reduce interference between multiple End devices, the ping slot is randomly changed in each beacon period, as illustrated in Fig. 1. In the example, End device 1 and End device 2 have the same periodicity factor $k = 1$. Therefore, both devices open two ping slots spaced 2048 slots apart, but with different offsets. A gateway sends a downlink packet to each End device. The shift between the ping slots of the two end nodes changes at each beacon period.

B. Energy Consumption of Class-A and Class-B Modes

Beacons reception allows an End device to adjust its internal clock, improving synchronization—an essential requirement for critical monitoring systems. However, the need to open multiple reception windows in Class-B mode significantly increases energy consumption, potentially shortening battery

life compared to Class-A mode. To highlight this tradeoff, we now provide a power consumption analysis of both modes.

As battery-powered devices operate at constant voltage, their average power consumption is proportional to the average current drawn across the different states through which the device transitions. For a Class-A device, the reference operational cycle includes only three states: Sleep (S), Transmission (T_x), and Reception (R_x). We therefore denote by I_S , I_{T_x} , and I_{R_x} the current absorbed in each of these states.¹ Moreover, we indicate by $T_{T_x}^U$ and $T_{T_x}^D$ the *time-on-air* of the uplink and downlink packet transmissions, respectively, and by T_{R_x} the standard duration of each of the two reception windows opened after each transmission, in case no downlink packet is detected. Note that if a downlink packet is received in any of the reception windows, the reception phase will last until the end of packet transmission, that is, for a time $T_{T_x}^D > T_{R_x}$. In the considered scenario, downlink packet transmission is rather sporadic, therefore this event does not practically impact on the energy efficiency of the system. However, for the consistency of the model, we prefer to include it in our analysis, considering the probability $p_{R_x}^D$ that a downlink packet is transmitted in a certain operational cycle.

Let T_{Cyc} denote the duration of a sleep-transmission-reception cycle. The average current absorbed in a such a cycle can hence be estimated as

$$I_A = \frac{I_S T_S + I_{T_x} T_{T_x}^U + 2I_{R_x} T_{R_x} + p_{R_x}^D I_{R_x} (T_{T_x}^D - T_{R_x})}{T_{Cyc}} \quad (1)$$

where $T_S = T_{Cyc} - T_{T_x} - 2T_{R_x} - p_{R_x}^D (T_{T_x}^D - T_{R_x})$ is the average time in a cycle interval during which the device is neither transmitting nor receiving.

On the other hand, in Class-B mode, a device wakes up also to receive the beacons and listen to the ping slots. For a fair comparison, we consider the same packet transmission events in a cycle, for both the uplink and the downlink. Moreover, we assume that T_{Cyc} contains an integer number of beacon periods of duration T_{BCN} . Let $T_{R_x}^{bcn}$ and $T_{R_x}^{png}$ be the duration of the reception windows for the beacon and a ping slot, respectively. Then, the average current absorbed in a time interval T_{Cyc} by a Class-B device can be expressed as

$$I_B = I_A + I_{PNG} + I_{BCN}, \quad (2)$$

where I_A is as in (1), whereas

$$I_{PNG} = \frac{(I_{R_x} - I_S)}{T_{BCN}} T_{R_x}^{png} 2^k \quad (3)$$

and

$$I_{BCN} = \frac{(I_{R_x} - I_S)}{T_{BCN}} T_{R_x}^{bcn} \quad (4)$$

represent the extra average current required to open the ping slots and receive the beacons, respectively.

Computing the ratio between the energy efficiencies of Class-B and Class-A modes we get

¹We neglect any variations in the absorbed current during state transitions. These are generally negligible in terms of overall energy consumption.

$$\rho = \frac{I_B}{I_A} = 1 + \frac{T_{Cyc}}{T_{BCN}} (I_{R_x} - I_S) \left(T_{R_x}^{png} 2^k + T_{R_x}^{bcn} \right) \cdot \left(I_S T_S + I_{T_x} T_{T_x}^U + 2I_{R_x} T_{R_x} + p_{R_x}^D I_{R_x} (T_{T_x}^D - T_{R_x}) \right)^{-1}. \quad (5)$$

By setting the model's parameters to their actual values, we obtain the extra power consumption of Class-B mode with respect to the basic Class-A mode. Some typical values for commonly used hardware are the following [36]: $I_S = 1 \mu A$, $I_{T_x} = 87 \text{ mA}$, and $I_{R_x} = 12 \text{ mA}$. Considering a cycle period of 30 min, and setting $k = 1$, and $p_{R_x}^D = 1/48 \simeq 0.02$, corresponding to one downlink packet transmission per day, on average, we obtain

$$\rho = 1 + 0.77 \frac{T_{Cyc}}{T_{BCN}} \simeq 11, 83. \quad (6)$$

Therefore, the extra power consumption of Class-B mode with respect to Class-A mode increases linearly with the ratio $\mu = (T_{Cyc}/T_{BCN})$ that represents the number of beacon periods within a cycle, i.e., between consecutive transmissions of the device. This suggests that, in the scenario considered, a device operating in Class-A mode will last ten times longer than an identical device operating in Class-B mode, reflecting the energy cost of synchronization in LoRaWAN networks.

C. Energetic Impact of Beacon Period

Based on the previous energetic analysis, suppressing the reception of some beacons may potentially improve the energy efficiency of a Class-B device. To substantiate this intuition, we compute the expected lifetime of a Class-B device powered by a 1500 mAh battery when only one beacon every $\beta \in \mathbb{N}$ is received. In other words, we consider a *stretched* beacon period $T'_{BCN} = \beta T_{BCN}$, where $T_{BCN} = 128 \text{ s} \simeq 2.13 \text{ min}$ is the standard beacon duration. The average current absorbed for beacon reception thus becomes $I'_{BCN} = (I_{BCN}/\beta)$, where I_{BCN} is given in (2), whereas the other terms remain unaffected by β . In particular, I_{PNG} is not varied because the number of ping slots per unit of time remains unchanged to ensure full adherence to the LoRaWAN standard.

The results are reported in Fig. 2, which shows how the battery lifetime increases significantly with β , i.e., with the *stretched* beacon period. This effect is particularly notable when the number of ping slots is small, a common scenario in IoT-based applications [35]. Thus, although the number of ping slots significantly affects the overall energy efficiency of the device, extending the beacon period can still greatly increase battery lifespan.

IV. SYSTEM MODEL

As previously discussed, beacon-based synchronization systems incur an unavoidable baseline energy cost proportional to the beacon reception rate: the higher the number of beacons received in a certain period, the higher the energy cost due to synchronization. However, reducing the beacon reception rate can lead to synchronization problems, as the accumulated clock drift might exceed the time accuracy required by the monitoring application. Therefore, there is a tradeoff between energy efficiency and time synchronization accuracy.

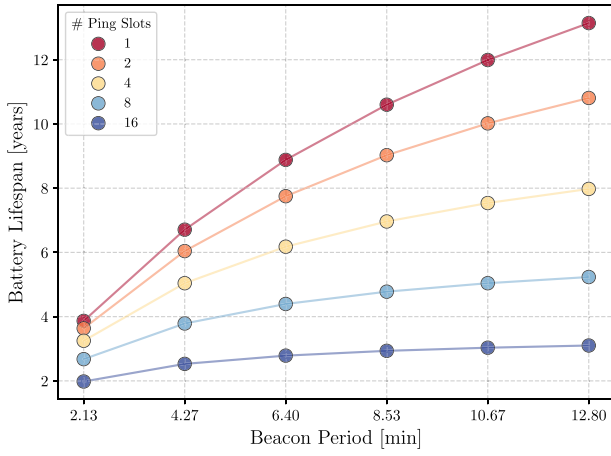


Fig. 2. Battery life of an End device transmitting 16-byte uplink packets every 30 min at $SF = 7$ and receiving one 16-byte downlink per day, with a battery capacity of 1500mAh.

To improve this tradeoff, an End device can estimate its clock drift and compensate for it. In this way, the device can save energy by skipping beacon receptions as long as the time misalignment remains within an acceptable range. To formally define a mechanism for such a prediction, we first introduce some notation. Table I summarizes the main symbols used in the following sections for the reader's convenience.

We indicate absolute time by t . For a generic device i with tick period τ_i , we define the *normalized* time as

$$t_i \stackrel{\text{def}}{=} t/\tau_i \quad (7)$$

and its *discretized* version, referred to as the *clock time*, as

$$\bar{t}_i \stackrel{\text{def}}{=} \lfloor t_i \rfloor, \quad (8)$$

where $\lfloor \cdot \rfloor$ indicates the *integer part* operator. In fact, $\bar{t}_i$ defines the integer count of tick periods measured by device i in a time interval of duration t . Moreover, we define the discretization error as $r_i \stackrel{\text{def}}{=} t_i - \bar{t}_i$, with $r_i \in [0, 1)$. We assume that the time reference is provided by the clock of the gateway that we indicate as device 0.

Now, let $t^{(n)}$ be the absolute time at which the n th beacon is sent by the gateway. Therefore, $t_i^{(n)}$ is the normalized time at which the n th beacon is received by the i th device, and $\bar{t}_i^{(n)}$ is the corresponding tick count. Let M be the nominal duration of the beacon period, expressed in number of clock ticks of the reference device, so that $t_0^{(n)} = \bar{t}_0^{(n)} = nM$ for any $n \in \mathbb{N}$. The parameter M is known by all devices within the network. However, the actual duration of the beacon interval measured by other devices may differ from M because their tick times differ from τ_0 . We then define the n th *beacon period offset* of device i as

$$\phi_i^{(n)} \stackrel{\text{def}}{=} t_i^{(n+1)} - t_i^{(n)} - M. \quad (9)$$

Since $\phi_i^{(n)}$ takes values in the real set, its exact value is inaccessible to the device that can only measure the tick counts between the two beacon events, and subtract M from

it. Therefore, the *measurable* n th beacon period offset is given by

$$\tilde{\phi}_i^{(n)} \stackrel{\text{def}}{=} \bar{t}_i^{(n+1)} - \bar{t}_i^{(n)} - M.$$

By replacing the discretization errors in (9), we have

$$\begin{aligned} \phi_i^{(n)} &= \bar{t}_i^{(n+1)} - \bar{t}_i^{(n)} - M + R_i^{(n)} \\ &= \tilde{\phi}_i^{(n)} + R_i^{(n)} \end{aligned} \quad (10)$$

where $R_i^{(n)} \stackrel{\text{def}}{=} r_i^{(n+1)} - r_i^{(n)}$ is named *residual value*. It should be noted that $R_i^{(n)} \in (-1, 1)$, so that the measurable offset $\tilde{\phi}_i^{(n)}$ can differ from $\bar{\phi}_i^{(n)} \stackrel{\text{def}}{=} \lfloor \phi_i^{(n)} \rfloor$ by at most $+1$ or -1 , depending on the sign and value of $R_i^{(n)}$. Time displacements smaller than a tick period accumulate over consecutive beacon periods, eventually resulting in a variation $+1$ or -1 of the measurable beacon period offset $\tilde{\phi}_i^{(n)}$ any time $r_i^{(n)}$ resets.

Fig. 3 illustrates a scenario in which an End device receives beacons from a gateway in an environment where the beacon period offset is constant and equal to $\phi_i^{(n)} = 2.66$ tick units. In this case, the End device's clock runs faster than the reference clock, resulting in additional ticks being added during each beacon period, with the measurable offset $\tilde{\phi}_i^{(n)}$ taking values in the set $\{2, 3\}$.

Observing the variation pattern of $\tilde{\phi}_i^{(n)}$, a device can attempt to estimate the actual drift of its clock and correct it so as to skip the reception of some beacons while keeping clocks aligned with the reference device. To better explain the idea, consider a node that at time $t^{(n)}$ attempts to predict the reception time of the next k beacons. The $(n+k)$ th normalized beacon reception time for device i is given by

$$t_i^{(n+k)} = t_i^{(n)} + kM + \sum_{l=0}^{k-1} \phi_i^{(n+l)}. \quad (11)$$

However, the actual clock time of the $(n+k)$ th beacon is given by

$$\bar{t}_i^{(n+k)} = \bar{t}_i^{(n)} + kM + \sum_{l=0}^{k-1} \tilde{\phi}_i^{(n+l)}. \quad (12)$$

The node then needs to estimate the k offset values to be added to its own counting of the M tick periods (corresponding to a nominal beacon interval) to keep in synchronization with device 0. Then, let $\hat{\phi}_i^{(n+l;n)}$ be the estimate of the offset for the next l th beacon reception time, with $l \in \{0, 1, \dots, k-1\}$. The estimated clock time of the $(n+k)$ th beacon, given the current clock time $\bar{t}_i^{(n)}$, will be

$$\hat{\bar{t}}_i^{(n+k;n)} \stackrel{\text{def}}{=} \bar{t}_i^{(n)} + kM + \sum_{l=0}^{k-1} \hat{\phi}_i^{(n+l;n)}. \quad (13)$$

The total error associated with this estimate—hereafter referred to as the *accumulated error*, as it reflects the cumulative deviation over k estimation steps—is formally defined as

$$\epsilon_{\hat{\phi}_i^{(n+k;n)}}^{\text{acc}} \stackrel{\text{def}}{=} \bar{t}_i^{(n+k)} - \hat{\bar{t}}_i^{(n+k;n)} = \sum_{l=0}^{k-1} (\tilde{\phi}_i^{(n+l)} - \hat{\phi}_i^{(n+l;n)}). \quad (14)$$

MAIN NOTATION AND DEFINITIONS

Symbol Definition	Description
t	Absolute time.
τ_i	Tick period of device i 's crystal oscillator.
$f_i = 1/\tau_i$	Oscillation frequency of device i 's crystal oscillator.
$t^{(n)}$	Arrival time of the n -th beacon; $n \in \mathbb{N}$.
$\Delta t^{(n)} = t^{(n+1)} - t^{(n)}$	Duration of the n -th beacon period.
$t_i = \frac{t}{\tau_i}$	Absolute time t normalized by the tick period of device i ; $t_i \in \mathbb{R}_{>0}$.
$t_i^{(n)} = t^{(n)}/\tau_i$	Arrival time of the n -th beacon normalized by the tick period of device i ; $t_i^{(n)} \in \mathbb{R}_{>0}$.
$\tilde{t}_i^{(n)} = \lfloor t_i^{(n)} \rfloor$	Number of full ticks counted by device i at time $t^{(n)}$; $\tilde{t}_i^{(n)} \in \mathbb{N}$.
$\phi_i^{(n)} = t_i^{(n+1)} - t_i^{(n)} - M$	Offset of the n -th beacon period relative to the nominal value; $\phi_i^{(n)} \in \mathbb{R}$.
$\tilde{\phi}_i^{(n)} = \tilde{t}^{(n+1)} - \tilde{t}^{(n)} - M$	Measurable version of $\phi_i^{(n)}$; $\tilde{\phi}_i^{(n)} \in \mathbb{Z}$.
$R_i^{(n)} = r_i^{(n+1)} - r_i^{(n)}$	Residual value; $R_i^{(n)} \in (-1, 1)$.
$\hat{t}_i^{(n+k;n)}$	Estimate of $\tilde{t}_i^{(n+k)}$ given $\tilde{t}_i^{(n)}$; $\hat{t}_i^{(n+k;n)} \in \mathbb{N}$.
$\hat{\phi}_i^{(n+k;n)}$	Estimate of $\tilde{\phi}_i^{(n+k)}$ given $\tilde{\phi}_i^{(n)}$; $\hat{\phi}_i^{(n+k;n)} \in \mathbb{Z}$.
$T(t)$	Temperature at time t .
$T^{(n)} = T(t^{(n)})$	Temperature at the n -th beacon arrival time.
$\Delta T^{(n)} = T^{(n+1)} - T^{(n)}$	Temperature change during the n -th beacon period.
$\delta T^{(k,n)} = T^{(n+k)} - T^{(n)}$	Temperature change over k beacon periods starting from $t^{(n)}$.
$\alpha_i^{(n+k,n)} = \phi_i^{(n+k)} - \phi_i^{(n)}$	Beacon period offset change between the n -th and $(n+k)$ -th beacon periods; $\alpha_i^{(n+k,n)} \in \mathbb{R}$.
$\hat{\alpha}_i^{(n+k;n)}$	Estimate of $\alpha_i^{(n+k,n)}$ given $\tilde{\phi}_i^{(n)}$ and $\Delta T^{(n)}$; $\hat{\alpha}_i^{(n+k;n)} \in \mathbb{R}$.

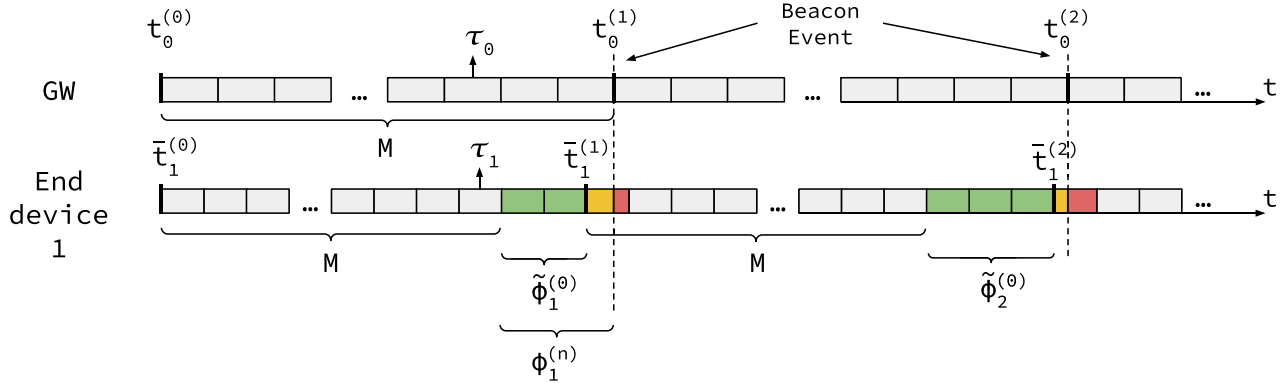


Fig. 3. Diagram illustrates the time evolution of tick counts on device 0 (e.g., the gateway) and device 1 during the transmission and reception of beacons. In this particular case, $\tau_1 < \tau_0$, meaning that over a beacon period, device 1 registers more ticks than the nominal value M . The measured beacon period offset is shown in green. As this example assumes a constant $\phi_1^{(n)}$ with a value of 2.66 tick units, the measurable offset $\tilde{\phi}_1^{(n)}$ equals 2 ticks at the first beacon reception ($n = 1$), and 3 ticks at the second ($n = 2$). This shows that $\tilde{\phi}_i$ is not always necessarily equal to $\tilde{\phi}_i$ (in this case, $\tilde{\phi}_i = \lfloor \phi_i \rfloor = 2$). The discretization value $\tau_i^{(n)}$ is shown in yellow, while the remaining fraction of the tick is shown in red. Together, these two fractional components represent the first tick counted by the device in the subsequent beacon period.

In this last expression, each term within the summation captures the discrepancy between the true and estimated offset values at a single prediction step. We denote these individual discrepancies as

$$\epsilon_{\hat{\phi}_i^{(n+l;n)}} \stackrel{\text{def}}{=} \tilde{\phi}_i^{(n+l)} - \hat{\phi}_i^{(n+l;n)} \quad (15)$$

and refer to each of them as the *step error*. Both the step error and accumulated error are fundamental metrics to assess prediction performance. Moreover, the accumulated error is critically important as it is the only metric observable by End devices when a beacon is received.

V. EFFICIENT STRATEGY FOR SYNCHRONIZATION

In the following, we propose a method to obtain the set of estimates $\{\hat{\phi}_i\}$ that are required to predict clock deviations when using (13). We start by considering a simplified case

where environmental conditions remain constant, and then extend the analysis to a more realistic scenario where the conditions vary over time.

A. Error Prediction in Constant Environmental Conditions

Let us first consider a scenario in which device i exhibits a constant beacon period offset $\phi_i = -3.6$ relative to the gateway, *i.e.*, the clock of device i is slower than the reference. Let $\Delta t^{(n)} \stackrel{\text{def}}{=} t^{(n+1)} - t^{(n)}$ define the n th beacon period. Thus, upon receiving the first beacon, device i measures the offset $\tilde{\phi}_i^{(0)} = -4$ during $\Delta t^{(0)}$, assuming $t^{(0)} = 0$ without loss of generality. It then updates the tick counter and enters sleep mode until the next beacon arrival. At this point, the discretization error results $r_i^{(0)} = 0.4$. When the following beacon arrives, the device wakes up and measures the beacon period offset during $\Delta t^{(1)}$, resulting in $\tilde{\phi}_i^{(1)} = -4$ with $r_i^{(1)} =$

TABLE II
EXAMPLE OF DEVICE i 'S CLOCK TICK EVOLUTION FOR A BEACON
PERIOD OFFSET $\phi_i^{(n)} = -3.6$ AND $M = 100$

l	0	1	2	3	4
$t_i^{(l)}$	96.4	192.8	289.2	385.6	482
$r_i^{(l)}$	0.4	0.8	0.2	0.6	0
$\tilde{\phi}_i^{(l)}$	-4	-4	-3	-4	-3
$R_i^{(l)}$	0.4	0.4	-0.6	0.4	-0.6

0.8. Table II summarizes the remaining sequence of steps until $r_i^{(n)}$ reaches zero and the cycle restarts.

For this specific case, the *correction pattern*, represented by the row $\{\tilde{\phi}_i^{(l)}\}$ in Table II, is finite. Thus, after receiving the first five beacons, device i can start predicting using (13), with the prediction error guaranteed to be exactly zero every five predicted steps. However, this behavior depends entirely on the value of ϕ_i . In fact, for certain values, the pattern may be infinite and not cyclic, making it impractical for a real-world implementation.

To address this limitation, we leverage the binary nature of the sequence $\tilde{\phi}_i^{(l)}$, which enables the use of binary decoding techniques to estimate $\phi_i^{(n)}$ from the observed $\tilde{\phi}_i^{(l)}$ values. The resulting estimate can then be used to reconstruct the pattern sequence and compute (13). However, if τ_i varies due to external environmental factors, the pattern becomes variable, and so a more versatile mechanism must be applied. In the following, we propose a possible approach for this case.

B. Error Prediction in Changing Environmental Conditions

In IoT applications where environmental factors fluctuate, τ_i , and therefore f_i , become time-dependent variables, denoted as $\tau_i(t)$ and $f_i(t)$, respectively. To analyze their impact on the time measured by an End device, we start by considering a time interval of duration t , divided into m subintervals of (potentially uneven) durations $\{t'_1, t'_2, \dots, t'_m\}$. During each subinterval, we assume that the clock period remains constant. Under this assumption, the ticks counted by device i can be modeled as a monotonically increasing function of t

$$t_i = \sum_{p=1}^m \frac{t'_p}{\tau_i(t'_p)}, \quad (16)$$

where $\tau_i(t'_p)$ represents the clock tick during the p th subinterval. As the durations of the subintervals t'_p become arbitrarily small, in the limit, the summation becomes an integral given by

$$t_i = \int_0^t \frac{1}{\tau_i(t)} dt. \quad (17)$$

Alternatively, this can be expressed in terms of $f_i(t)$ as

$$t_i = \int_0^t f_i(t) dt. \quad (18)$$

While factors, such as temperature, humidity, pressure, and power instability can affect the instantaneous frequency $f_i(t)$, our model mainly focuses on temperature and aging, as the

influence of the remaining factors becomes negligible when modern industrial and electrical design practices are applied.

Numerous studies have indicated that the frequency-temperature dependency of tuning fork crystals, which are the ones commonly used in low-power IoT devices, exhibits a parabolic behavior, mainly due to the type of cut employed [37], [38], [39]. This relationship is often described by the function

$$f_i(t) = f_0(M_0 + \gamma(T(t) - T_0)^2) \quad (19)$$

where f_0 represents the nominal frequency, M_0 the frequency tolerance, T_0 is the turn-over temperature, and γ is a constant known as *temperature coefficient*. The temperature coefficient, expressed in units of ppm/°C², quantifies the influence of temperature on the crystal drift. Thus, by substituting (19) into (18), we obtain

$$t_i = \int_0^t f_0(M_0 + \gamma(T(t) - T_0)^2) dt. \quad (20)$$

The temperature $T(t)$ is, at first glance, an unknown function. That said, it is important to note that, in the IoT-based monitoring applications under consideration, temperature does not change suddenly; rather, it might fluctuate but on a timescale of tens or hundreds of seconds. Consequently, it is reasonable to assume that if we consider a sufficiently small time interval $\Delta t^{(n)}$, temperature variations can be approximated by a first-order polynomial. Based on this assumption, the temperature can be expressed as follows:

$$T(t) = T^{(n)} + \frac{\Delta T^{(n)}}{\Delta t^{(n)}} t$$

with $t \in [0, \Delta t^{(n)}]$, where $T^{(n)}$ represents the temperature at time $t^{(n)}$ and $\Delta T^{(n)} \stackrel{\text{def}}{=} T^{(n+1)} - T^{(n)}$. Consequently, using (9), the beacon period offset during $\Delta t^{(n)}$ becomes

$$\begin{aligned} \phi_i^{(n)} &= \int_0^{t^{(n+1)}} f_i(t) dt - \int_0^{t^{(n)}} f_i(t) dt - M \\ &= f_0 M_0 \Delta t^{(n)} \\ &\quad + f_0 \gamma \int_0^{\Delta t^{(n)}} (a^2 + 2abt + (bt)^2) dt - M \end{aligned} \quad (21)$$

where $a = T^{(n)} - T_0$ and $b = [\Delta T^{(n)} / \Delta t^{(n)}]$. Thus, to estimate the offset during the interval $\Delta t^{(n+k)}$ based on $\phi_i^{(n)}$, we first introduce the *difference factor* α_i . This factor quantifies the beacon period offset change between the n th and $(n+k)$ th beacon periods and is defined as

$$\alpha_i^{(n+k,n)} \stackrel{\text{def}}{=} \phi_i^{(n+k)} - \phi_i^{(n)}. \quad (22)$$

Then, under the assumption of linearity in $T(t)$, and given that $\Delta t^{(n+k)} = \Delta t^{(n)} = \Delta t$, by using (21), the difference factor can be further expressed as follows:

$$\begin{aligned} \alpha_i^{(n+k,n)} &= \int_0^{\Delta t^{(n+k)}} f_0(M_0 + \gamma'(T(t)^{(n+k)} - T_0')^2) dt \\ &\quad - \int_0^{\Delta t^{(n)}} f_0(M_0 + \gamma'(T(t)^{(n)} - T_0')^2) dt \end{aligned}$$

$$\begin{aligned}
&= f_0 \gamma' \left[\left((T^{(n+k)} - T'_0)^2 - (T^{(n)} - T'_0)^2 \right) \right. \\
&\quad \left. + \left(\Delta T^{(n+k)} (T^{(n+k)} - T'_0) - \Delta T^{(n)} (T^{(n)} - T'_0) \right) \right. \\
&\quad \left. + \frac{1}{3} \left((\Delta T^{(n+k)})^2 - (\Delta T^{(n)})^2 \right) \right] \Delta t \\
&= f_0 \gamma' D'_{k,n} \Delta t \tag{23}
\end{aligned}$$

where $T'_0 = T_0(1 + e_{T_0})$, $\gamma' = \gamma(1 + e_\gamma)$, and $D'_{k,n} = D_{k,n} + e_{D_{k,n}}$, with

$$\begin{aligned}
D_{k,n} &= (T^{(n+k)})^2 - (T^{(n)})^2 + 2T_0(T^{(n)} - T^{(n+k)}) \\
&\quad + (\Delta T^{(n+k)}(T^{(n+k)} - T_0) - \Delta T^{(n)}(T^{(n)} - T_0)) \\
&\quad + \frac{1}{3} \left((\Delta T^{(n+k)})^2 - (\Delta T^{(n)})^2 \right)
\end{aligned}$$

and

$$e_{D_{k,n}} = e_{T_0} T_0 \left(2(T^{(n)} - T^{(n+k)}) + \Delta T^{(n)} - \Delta T^{(n+k)} \right).$$

Here, the values e_γ and e_{T_0} represent the deviations of the actual parameters γ' and T'_0 from their nominal values γ and T_0 , respectively. The precise determination of e_γ and e_{T_0} requires an individual calibration process for each crystal. However, crystal manufacturers also define tolerance limits for these parameters, which might be useful to evaluate worst-case scenarios.

Thus, in the absence of a calibration process, where only the nominal values γ and T_0 are available, we can formulate the estimate $\hat{\alpha}_i$ as

$$\hat{\alpha}_i^{(n+k;n)} = f_0 \gamma D_{k,n} \Delta t. \tag{24}$$

Then, the estimate's error is given by

$$\begin{aligned}
\epsilon_{\hat{\alpha}_i^{(n+k;n)}} &= \alpha_i^{(n+k;n)} - \hat{\alpha}_i^{(n+k;n)} \\
&= f_0 \gamma' D'_{k,n} \Delta t - f_0 \gamma D_{k,n} \Delta t \\
&= f_0 \gamma [e_\gamma (D_{k,n} + e_{D_{k,n}}) + e_{D_{k,n}}] \Delta t. \tag{25}
\end{aligned}$$

Having defined $\hat{\alpha}_i^{(n+k;n)}$ and its error, we can now proceed to define the estimate $\hat{\phi}_i$ to be used in (13). Thus, after k steps have elapsed since the reception of the n th beacon, based on (22), the estimate can be expressed as follows:

$$\hat{\phi}_i^{(n+k;n)} = \tilde{\phi}_i^{(n)} + \lfloor \hat{\alpha}_i^{(n+k;n)} \rfloor. \tag{26}$$

As (26) shows, the estimate $\hat{\phi}_i^{(n+k;n)}$ relies on the most recent measurement that took place at $t^{(n)}$. Therefore, as aging effects manifest over much longer timescales, their influence is negligible within the tens-of-minutes window considered, and thus they do not affect the prediction.

Using (15), the *step error* of this new estimate becomes

$$\begin{aligned}
\epsilon_{\hat{\phi}_i^{(n+k;n)}} &= \tilde{\phi}_i^{(n+k;n)} - \hat{\phi}_i^{(n+k;n)} \\
&= \phi_i^{(n+k)} - R_i^{(n+k)} - \left(\tilde{\phi}_i^{(n)} + \lfloor \hat{\alpha}_i^{(n+k;n)} \rfloor \right) \\
&= \alpha_i^{(n+k;n)} - \lfloor \hat{\alpha}_i^{(n+k;n)} \rfloor - \left(R_i^{(n)} - R_i^{(n+k)} \right) \tag{27}
\end{aligned}$$

On the other hand, the *accumulated error* after the same k steps is given by (14).

From analyzing the different terms in (27), it can be deduced that, when $\alpha_i^{(n+k;n)} = \hat{\alpha}_i^{(n+k;n)}$, the expression in (27) can only take values in the set $\{-2, -1, 0, 1, 2\}$. Instead, when $\alpha_i^{(n+k;n)} \neq \hat{\alpha}_i^{(n+k;n)}$, the step error is unbounded. Additionally, it is important to note that the continuous temperature variations induce stochastic behaviors in each of the variables within (27), causing them to behave as random variables. As a result, $\epsilon_{\hat{\phi}_i^{(n+k;n)}}$ becomes another random variable with a specific distribution which will be further analyzed in the results section.

To study the influence of $D_{k,n}$ and $e_{D_{k,n}}$ on $\epsilon_{\hat{\alpha}_i^{(n+k;n)}}$, and subsequently on $\epsilon_{\hat{\phi}_i}$ and $\epsilon_{\hat{\phi}_i}^{\text{acc}}$, we first define the change in temperature after k steps as $\delta T^{(k,n)} \stackrel{\text{def}}{=} T^{(n+k)} - T^{(n)}$, which allows us to rewrite $D_{k,n}$ as

$$\begin{aligned}
D_{k,n} &= (\delta T^{(k,n)})^2 + 2\delta T^{(k,n)}(T^{(n)} - T_0) \\
&\quad + \Delta T^{(n+k)}(T^{(n)} + \delta T^{(k,n)} - T_0) \\
&\quad - \Delta T^{(n)}(T^{(n)} - T_0) \\
&\quad + \frac{1}{3} \left((\Delta T^{(n+k)})^2 - (\Delta T^{(n)})^2 \right) \tag{28}
\end{aligned}$$

and $e_{D_{k,n}}$ as

$$e_{D_{k,n}} = e_{T_0} T_0 \left(-2\delta T^{(k,n)} + \Delta T^{(n)} - \Delta T^{(n+k)} \right). \tag{29}$$

As, typically, $e_\gamma \in [-0.111, 0.111]$ and $e_{T_0} \in [-0.2, 0.2]$ [39], we can compute $D_{k,n}$, $e_{D_{k,n}}$, and, consequently, $\epsilon_{\hat{\alpha}_i^{(n+k;n)}}$ under worst-case scenarios. In a LoRaWAN Class-B environment, where $\Delta t = 128$ s, it is reasonable to set temperature limits, such as $\Delta T_{\text{max}} = 0.5^\circ\text{C}$ and $\delta T_{\text{max}} = 5^\circ\text{C}$ to ensure system stability and prevent unexpected peaks in $\epsilon_{\hat{\phi}_i}^{\text{acc}}$.² Under these constraints, and considering the crystal's operational temperature range from -40°C to 80°C , $|D_{k,n}|$ reaches its maximum value, $D_{k_{\text{max}},n}^* = 742.5^\circ\text{C}^2$, when $k = k_{\text{max}}$, $\Delta T^{(n)} = -\Delta T^{(n+k_{\text{max}})} = \Delta T_{\text{max}}$, $\delta T^{(k_{\text{max}},n)} = -\delta T_{\text{max}}$, and $T^{(n)} = -40^\circ\text{C}$. Fig. 4 shows the behavior of $D_{k,n}$ as a function of the temperature $T^{(n)}$ for different $\delta T^{(k,n)}$ configurations under worst-case conditions, i.e., $\Delta T^{(n)} = -\Delta T^{(n+k)} = \Delta T_{\text{max}}$. The strong dependency of $D_{k,n}$, and also $e_{D_{k,n}}$, on $T^{(n)}$ and $\delta T^{(k,n)}$ suggests that thresholds does not necessarily need to remain static. Instead, they could dynamically adapt based on the operating temperature of the device, thereby further restricting predictions when errors are expected to grow more rapidly.

Thus, given $D_{k_{\text{max}},n}^*$, the estimate error results $\epsilon_{\hat{\alpha}_i^{(n+k_{\text{max}},n)}}^* = -21.6$. This occurs for a crystal with parameters $e_\gamma = 0.111$ and $e_{T_0} = 0.2$. In this specific case, $|\epsilon_{\hat{\phi}_i}|$ is also maximized, resulting in a step error $\epsilon_{\hat{\phi}_i^{(n+k_{\text{max}},n)}}^* = -23$. An error of 23 clock ticks for a 32 kHz crystal corresponds to a

²When the system predicts, all three parameters $\Delta T^{(n)}$, $\Delta T^{(n+k)}$, and $\delta T^{(k,n)}$ must be constrained such that $|\Delta T^{(n)}| \leq \Delta T_{\text{max}}$, $|\Delta T^{(n+k)}| \leq \Delta T_{\text{max}}$, and $|\delta T^{(k,n)}| \leq \delta T_{\text{max}}$. This helps maintain the system's operation within a linear regime of temperature changes during Δt . Together, such limits determine the maximum number of steps before receiving a new beacon, e.g., k_{max} , which is given by $k_{\text{max}} = \delta T_{\text{max}} / \Delta T_{\text{max}}$.

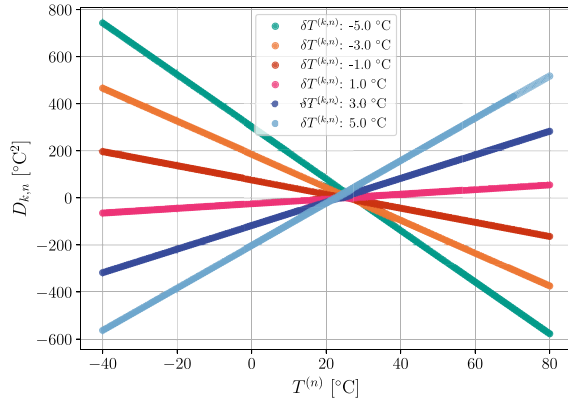


Fig. 4. Dependence of $D_{k,n}$ on the temperature $T^{(n)}$, computed using expression (28) for $T_0 = 25^\circ\text{C}$ and $\Delta T^{(n)} = -\Delta T^{(n+k)} = 0.5^\circ\text{C}$.

$702\mu\text{s}$ offset relative to the time source.³ Therefore, all the system thresholds must be carefully chosen, depending on the specific synchronization requirements of the application, ensuring that step and accumulated errors do not exceed the required limits.

At this point, we conclude that designing an algorithm based on the estimate $\hat{\alpha}_i$ requires measuring the beacon period offset at a given step, to then iteratively increase k while using (26) and (13) to predict clock drift in future steps. This adaptive strategy enables the mechanism to self-adjust during operation, thereby minimizing prediction errors. In the following, we introduce an algorithm based on the previously outlined model.

C. Algorithm

The algorithm presented in Algorithm 1 can be structured into three main blocks, which are iteratively repeated. In the first block, the device measures the temperature $T^{(n+j)}$ as the time for the next beacon reception approaches. In case a look-up table is employed to determine the parameter thresholds, the device subsequently calls the subfunction in Algorithm 2 to set them, and checks whether the temperature changes $\Delta T^{(n+j)}$ or $\delta T^{(n+j)}$ exceed the limits $\Delta T_{\max}$ or $\delta T_{\max}$, respectively.

The second block involves estimating the clock drift at the j th step based on the measured temperature using (26) when calling Algorithm 3. Then, the correction is computed and applied as in (13).

In the third block, the device determines whether to receive a beacon. If it is time to receive one, the End device measures the accumulated error $\epsilon_{\hat{\phi}_i}^{\text{acc}(n+j;n)}$. Based on the predefined threshold $\epsilon_{\hat{\phi}_i}^{\text{acc,max}}$, the algorithm updates k either by incrementing or decrementing its value. Once k is updated, the prediction cycle restarts. If it is not yet time to receive a beacon, the algorithm immediately restarts with the first block. A high-level illustration of the detailed algorithm is presented in the flowchart from Fig. 5.

³This specific scenario is highly unlikely. For such a level of step error to occur, the crystal parameters would need to be at their extreme tolerance limits, while a temperature change of $-\Delta T_{\max}$ would have to persist across all $k_{\max}$ steps. Furthermore, $\Delta T^{(n+k)}$ would need to equal $\Delta T_{\max}$ (i.e., a temperature change of opposite sign), and the operating temperature $T^{(n)}$ would have to be near its limit.

Algorithm 1 Adaptive Mechanism to Predict Corrections

```

1: Initialize  $k$ ,  $T$ ,  $\Delta T$ ,  $\delta T$ , and  $\hat{\phi}_i$ 
2: while true do
3:    $flag \leftarrow \text{false}$  ▷ clear restart flag
4:   for  $j \leftarrow 1$  to  $k$  do
5:     Wait  $T_{\text{BCN}}$  seconds
6:     Measure  $T^{(n+j)}$ 
7:     SETMAXVALUES( $T^{(n+j)}$ )
8:     Compute  $\Delta T^{(n+j)}$ 
9:     if  $\Delta T^{(n+j)} \geq \Delta T_{\max}$  or  $\delta T^{(n+j)} \geq \delta T_{\max}$  then
10:       $j \leftarrow k$ 
11:       $flag \leftarrow \text{true}$ 
12:    end if
13:     $\hat{\phi}_i^{(n+j;n)} \leftarrow \text{GETPREDICTION}(T^{(n+j)}, \Delta T^{(n+j)},$ 
       $T^{(n)}, \Delta T^{(n)}, \hat{\phi}_i^{(n)})$ 
14:    Apply prediction  $\hat{\phi}_i^{(n+j;n)}$  with (13)
15:    if  $j = k$  then ▷ A beacon must be received
16:      Do beacon reception
17:      Update  $T^{(n)}$ ,  $\Delta T^{(n)}$ , and  $\hat{\phi}_i^{(n)}$ 
18:      Compute  $\epsilon_{\hat{\phi}_i}^{\text{acc}(n+k;n)}$ 
19:      if  $\epsilon_{\hat{\phi}_i}^{\text{acc}(n+k;n)} > \epsilon_{\hat{\phi}_i}^{\text{acc,max}}$  or  $flag = \text{True}$  then
20:         $k \leftarrow \lfloor \frac{k}{2} \rfloor$  ▷ Reduction of  $k$ 
21:      else ▷ Increment of  $k$ , if feasible
22:        if  $k < k_{\max}$  then
23:           $k \leftarrow k + 1$ 
24:        else
25:           $k \leftarrow k_{\max}$ 
26:        end if
27:      end if
28:      break ▷ Exit the for loop to restart predicting
29:    end if
30:  end for
31: end while

```

Algorithm 2 Set Maximums Values

```

1: function SETMAXVALUES( $T^{(n+j)}$ )
2:   Use lookup table to set  $\Delta T_{\max}$  based on  $T^{(n+j)}$ 
3:   Use lookup table to set  $\epsilon_{\hat{\phi}_i}^{\text{acc,max}}$  based on  $T^{(n+j)}$ 
4:   Use lookup table to set  $\delta T_{\max}$  based on  $T^{(n+j)}$ 
5:    $k_{\max} \leftarrow \frac{\delta T_{\max}}{\Delta T_{\max}}$ 
6:   Return
7: end function

```

VI. EXPERIMENTAL SETTINGS

The entire firmware implementation deployed over the IoT devices was based on FreeRTOS, an open source real time system. The hardware primarily consisted of a Semtech SX1276 radio chipset paired with a *Wonder Gecko* ARM architecture micro-controller from Silicon Labs. The micro-controller was sourced by a 32 kHz low-energy crystal oscillator, resulting in a tick duration of $30.5\mu\text{s}$. To measure temperature, we used the micro-controller's built-in temperature sensor. Similarly, for the LoRaWAN gateway,

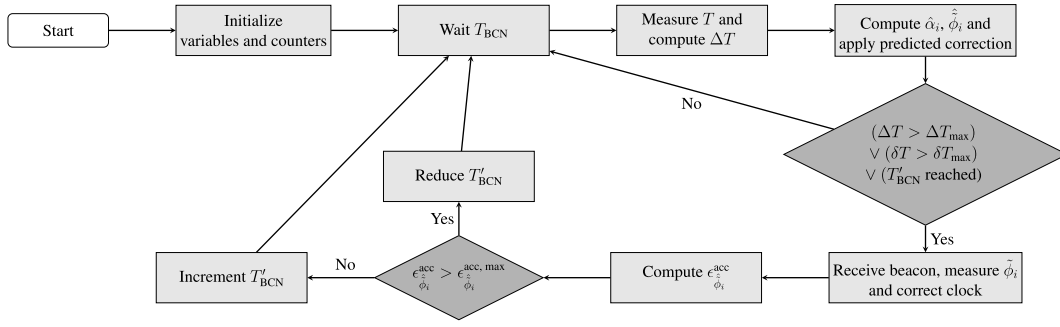


Fig. 5. High-level flowchart of the adaptive mechanism described in Section V-C. Here, T_{BCN} denotes the minimum prediction interval, which in our case is fixed by the LoRaWAN standard at 128 s. In contrast, T'_{BCN} is a *variable multiple* of T_{BCN} , representing the actual time between beacon receptions, as detailed in Section III-C.

Algorithm 3 Get Prediction Value

```

1: function GETPREDICTION( $T^{(n+j)}$ ,  $\Delta T^{(n+j)}$ ,  $T^{(n)}$ ,  $\Delta T^{(n)}$ ,  $\tilde{\phi}_i^{(n)}$ )
2:   Compute  $\hat{\alpha}_i^{(n+j;n)}$  using (24)
3:   Compute  $\hat{\phi}_i^{(n+j;n)}$  using (26)
4:   Return  $\hat{\phi}_i^{(n+j;n)}$ 
5: end function

```

a single-channel gateway with a similar radio chip was used. Implementing the adaptive mechanism on this hardware required a total of 2.5 kB of memory, making it feasible to install on most tiny micro-controllers used in IoT-based monitoring solutions.

To assess the system's performance under different conditions, we conducted four distinct experiments, focusing specially on two key performance indicators: the synchronization error and the average beacon period achieved by the devices. These metrics enabled us to assess the effectiveness of our solution in comparison with existing approaches.

The *first experiment* consisted on comparing the results obtained in a real environment with those from a simulated one. For this, End device 1 was placed outdoors over 90 h to measure temperature at each beacon period. During this phase, no adaptive mechanism was implemented; the primary goal was to collect a realistic temperature dataset, $T(t)$, which was later used to simulate the evolution of clock ticks using (20). At the same time, End device 2 was deployed outdoors with the adaptive mechanism active, which allowed for a comparison between the experimental and simulated results (an example of the deployment is shown in Fig. 6).

The *second experiment* aimed to evaluate how synchronization error and battery lifespan are influenced by configurable parameters. Lasting over 400 h, this experiment involved testing End device 2 with various values of $\Delta T_{\max}$ and $\epsilon^{\text{acc,max}}_{\phi_i}$. Subsequently, the *third experiment* was a long-term test designed to evaluate the robustness of the adaptive mechanism across multiple LoRa nodes. It involved End devices 2, 3, and 4, and lasted over 400 h. Finally, the *fourth experiment* was designed to evaluate extreme scenarios, focusing on the same metrics as in the previous experiments, but in cases where the temperature suddenly rises to very high levels.



Fig. 6. Outdoor deployment of an End device on the roof of a 100 m tall building in Barcelona, Spain.

VII. RESULTS AND DISCUSSION

We now present and analyze the results obtained from the experimental setups described earlier. As previously mentioned, our evaluation focuses on two metrics: 1) synchronization accuracy and 2) the average beacon period achieved by End devices. The former is quantified by the accumulated error $\epsilon^{\text{acc}}_{\phi_i}$, while the latter, represented by T'_{BCN} , reflects the average beacon period achieved throughout the experiments. The results presented in what follows aim to demonstrate the practical effectiveness of the proposed approach and its advantages over existing strategies.

A. Experiment 1: Simulation vs Experimental Results

1) *Simulations*: Once the temperature data was collected with device 1, the clock evolution was simulated by evaluating (20), with the assumption that linear temperature variations occurred between time steps Δt . To reduce noise and quantization errors, an *off-line* filter was applied to the raw data collected. Fig. 7 shows the temperature measurements before and after applying the filter, highlighting how the process improved data quality to support reliable simulation results.

Fig. 8(a) illustrates the results after applying the algorithm 1 to the synthetic clock tick evolution. The temperature profile is displayed at the bottom of the figure, while the top

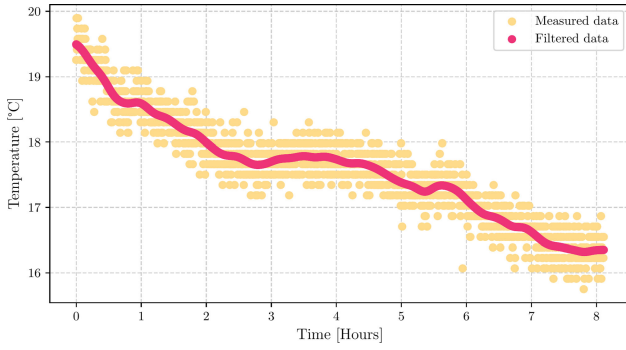


Fig. 7. Temperature readings from End device 1, both before and after the application of a Gaussian filter to mitigate noise and quantization errors.

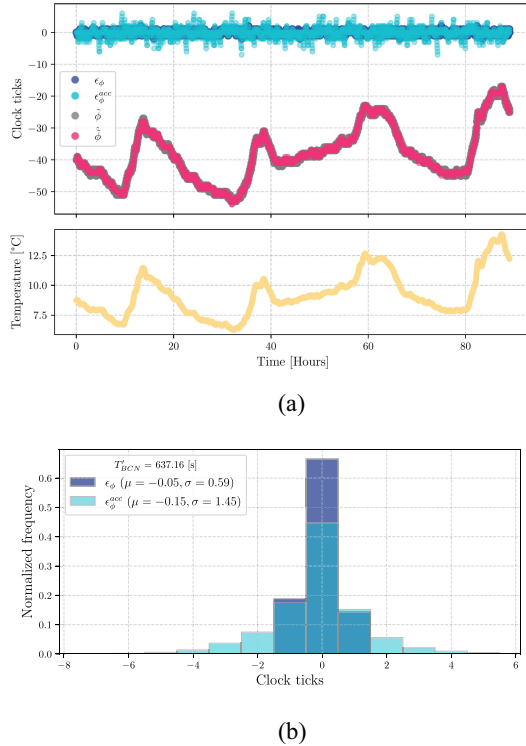


Fig. 8. Results obtained from applying Algorithm 1 to synthetic clock tick data generated from previously recorded temperature measurements. (a) Temporal evolution of $\tilde{\phi}_i$ and $\hat{\phi}$, along with the step and accumulated errors. At the bottom, the temperature readings after applying a Gaussian filter are displayed. (b) Histogram of the errors shown in Fig. 8(a).

section shows the temporal evolution of $\tilde{\phi}_i$ and $\hat{\phi}$, together with the step and accumulated errors.

As anticipated, the temperature dependence of the crystal frequency leads to a strong correlation between the corrections $\tilde{\phi}_i^{(n)}$ and temperature over time. In this case, since γ' and T'_0 are known, it follows that $\alpha_i^{(n+k,n)} = \hat{\alpha}_i^{(n+k,n)}$. Moreover, the temperature fluctuations introduce a stochastic component into the values in (27). As a result, these variables behave as uniformly distributed random variables. The subsequent mathematical operations in (27), which include addition, subtraction, and the integer operator $\lfloor \cdot \rfloor$, transform these individual stochastic contributions into a composite random

variable that exhibits a triangular distribution centered around zero.

Finally, since the accumulated error defined in (14) is the sum of k triangular distributions centered around zero, the resulting distribution exhibits a bell-shaped form that approaches a normal distribution as k increases, in accordance with the Central Limit Theorem. Fig. 8(b) presents the histogram of both errors, $\epsilon_{\phi_i}^{acc}$ and $\epsilon_{\phi_i}^{acc}$, where the observed triangular and bell-shaped distributions align perfectly with the theoretical distributions discussed above. From Fig. 8(b), we also see that the accumulated error, $\epsilon_{\phi_i}^{acc}$, reaches a maximum of -6 ticks, while the average beacon period exceeds 10 min. According to the estimations in Fig. 2, these results correspond to a system capable of extending the battery life by a factor of three compared to the standard Class-B configuration with 1 ping slot active, all while maintaining synchronization errors well below $200 \mu s$.

To study the impact of temperature measurement errors on $\epsilon_{\phi_i}^{acc}$ and $\epsilon_{\phi_i}^{acc}$, we proceed to analyze the distribution of the step error $\epsilon_{\phi_i}^{acc}$. The temperature measurement errors can be modeled by adding white noise to $T^{(n)}$, which introduces a new random variable, $D'_{k,n}$. Although γ' and T'_0 are known, $\epsilon_{\hat{\alpha}}$ becomes nonzero because now $\alpha_i^{(n+k,n)} = f_0 \gamma' D'_{k,n} \Delta t$, where $D'_{k,n} = D'_{k,n} + \sum_{j=0}^M \psi_j$, and each ψ_j is either a Gaussian distribution (zero mean, variance dependent on σ_T) or a symmetric bell-shaped distribution with faster decaying tails than a Gaussian. As a result, $\epsilon_{\hat{\alpha}_i}$ follows a bell-shaped distribution centered around zero. This matches the results in Fig. 9, where five different values of σ_T were tested. Consistently, the step error distribution becomes broader as σ_T increases. Similarly, the accumulated error, $\epsilon_{\phi_i}^{acc}$, also exhibits a widening trend given its dependence on the step error, reflecting the increased variability in temperature deviations.

2) *Experimental Results:* The results of the first implementation are shown in Fig. 10, where the devices ran the algorithm directly on board without any tick evolution simulation.⁴

As expected, the dispersion of both the step and the accumulated error that can be seen in Fig. 10(a) and (b) is higher than in the simulated case. This might be caused because of some temperature changes that are assumed to be lineal while they are not. Even though both γ' and T'_0 are not perfectly known, the values of α_i do not become excessively large because the system operates within nonextreme conditions. Specifically, the step error distribution follows a triangular shape, while the accumulated error tends to be Gaussian-shaped due to the Central Limit Theorem. As seen in the figures, the achieved beacon period is five times longer than the standard LoRaWAN Class-B beacon period of 128 s.

According to Fig. 9, temperature errors are expected to affect system performance. To mitigate this, the calibration procedures provided by the chip manufacturer were applied to

⁴It is worth noting that the End device used in the simulated clock evolution shown in Fig. 8 differs from the one used in the real measurements presented in Fig. 10, hence slightly different temperature measurements and clock offsets are expected.

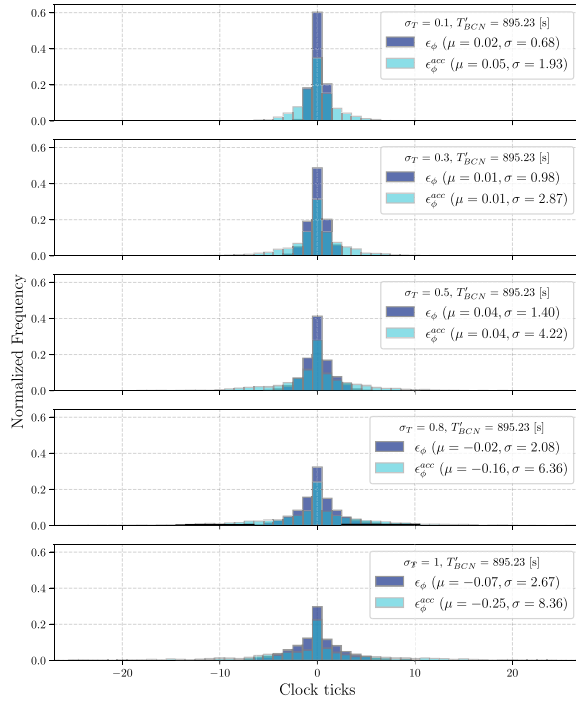


Fig. 9. Distribution of accumulated and step error obtained for a set of different temperature errors. To ensure a fair comparison, a fixed k , and thus a fixed T'_{BCN} , was used in all simulations.

improve sensor accuracy. Additionally, multiple temperature measurements were collected and averaged at each beacon period, further reducing the temperature uncertainty. To illustrate the impact of this averaging process, Fig. 11 presents results obtained without applying measurement averaging, in contrast to those shown in Fig. 10. When no averaging is applied, the error distribution $\epsilon_{\phi}^{\wedge}$ shifts from a triangular to a bell-shaped distribution, as observed in the simulations. Furthermore, the beacon period achieved by the algorithm was halved compared to the scenario with multiple temperature measurements, highlighting the importance of accurate temperature measurements.

B. Experiment 2: Parameter Dependencies

The second experiment aimed to assess how varying key parameters affects the behavior of the proposed mechanism. Specifically, five different values of $\Delta T_{\max}$ and three of $\epsilon_{\phi_i}^{\wedge, \max}$ were evaluated. As illustrated in Fig. 12, increasing $\Delta T_{\max}$ results in greater dispersion $\sigma_{\epsilon_{\phi_i}^{\wedge}}$, since the mechanism is allowed more flexibility in its predictions. This flexibility enables longer beacon intervals T'_{BCN} , thereby improving energy efficiency. In contrast, lower values of $\Delta T_{\max}$ impose stricter constraints on the predictions, leading to shorter T'_{BCN} intervals.

Next, by fixing a specific $\Delta T_{\max}$, we expect the step error to remain constant across different values of $\epsilon_{\phi_i}^{\wedge, \max}$. Since ΔT is checked at each step, both $D_{k,n}$ and $e_{D_{k,n}}$ are bounded, which in turn limits $\hat{a}_i^{(n+k;n)}$ at each step. Fig. 13 illustrates the histograms for $\Delta T_{\max} = 0.3$. Here, the step error

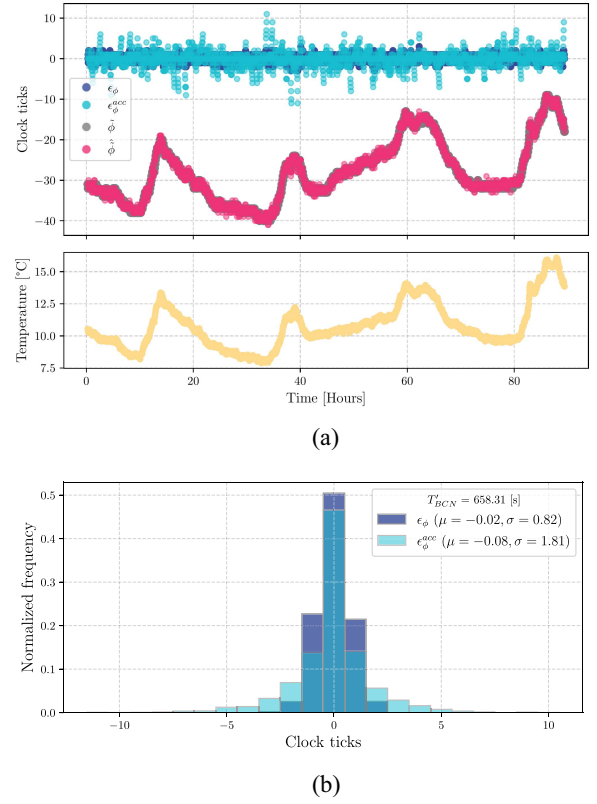


Fig. 10. Experimental results obtained with the adaptive mechanism proposed in Algorithm 1. Multiple temperature measurements were taken to compute an average, thereby minimizing the estimation error. (a) Time evolution of clock tick ground truth and predictions. (b) Histogram of step and accumulated errors.

maintains its triangular shape among different configurations. Naturally, a wider accumulated error is observed when a higher $\epsilon_{\phi_i}^{\wedge, \max}$ threshold is applied, as it allows k to increase more easily, thereby incorporating more terms into the sum in (14). Additionally, a higher $\epsilon_{\phi_i}^{\wedge, \max}$ is associated with a longer T'_{BCN} .

1) *Battery Lifespan Gain:* Using the same model employed for battery life estimation in Fig. 2, we assessed the potential battery life gains based on the experimentally measured beacon periods, T'_{BCN} , presented in Fig. 12. As illustrated in Fig. 14, battery life improvements decrease as the number of reception windows (*i.e.*, ping slots) increases, since the relative energy cost of beacon receptions becomes negligible compared to that of maintaining active reception windows. Nonetheless, in typical LoRaWAN monitoring deployments, where End devices commonly operate with no more than two ping slots due to predominantly uplink-oriented traffic, the proposed adaptive mechanism can effectively double or even triple battery lifespan of the standard case (*e.g.*, without dynamically adapting the beacon period).

C. Experiment 3: Long-Term Tests

The third set of experiments assessed the system's practicality through a long-term deployment. Fig. 15 displays the results, with each subfigure corresponding to a different

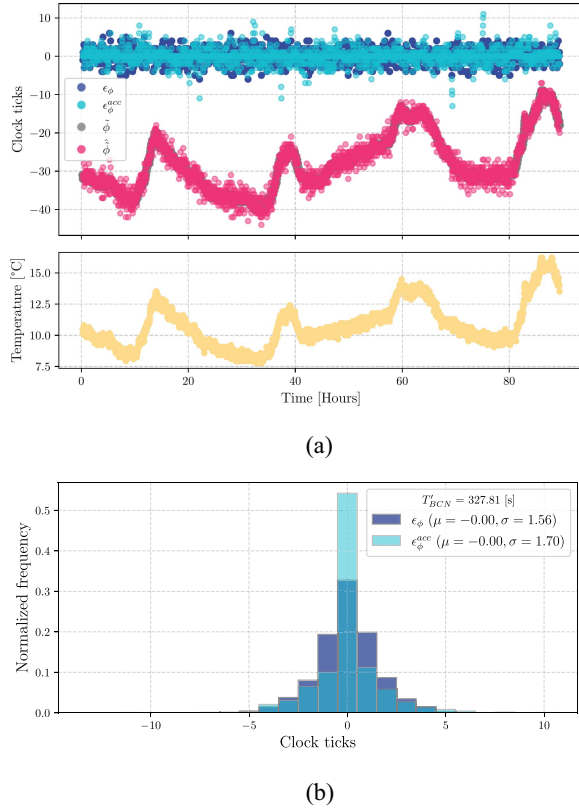


Fig. 11. Experimental results obtained with the same parameters as in Fig. 10, but without averaging temperature readings in the algorithm. (a) Time evolution of clock tick ground truth and predictions. (b) Histogram of the step and accumulated errors.

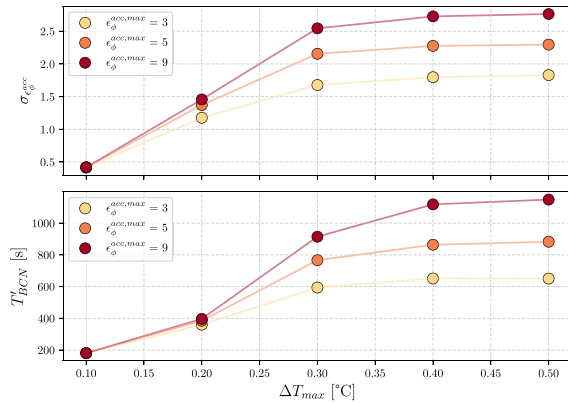


Fig. 12. System response to the thresholds $\Delta T_{\max}$ and $\epsilon_z^{\text{acc}, \max}$. The standard deviation of ϵ_z^{acc} is shown at the top, while the measured beacon period, T'_{BCN} , is displayed at the bottom.

hardware device. Variations in crystal characteristics led to differences in drift and average beacon periods across devices. The drift per period reached a maximum value of -42 ticks in device 2 and -66 ticks in device 4, highlighting the effects of manufacturing variability and hardware aging.

The accumulated error remained centered around zero for all devices, with occasional peaks of up to 15 clock-ticks. For End device 4, the mean beacon period offset was $\mu_{\hat{\phi}_i} = -38.1$. This indicates an average error accumulation of 38.1 clock ticks per step when no mechanism mitigates crystal drift. In

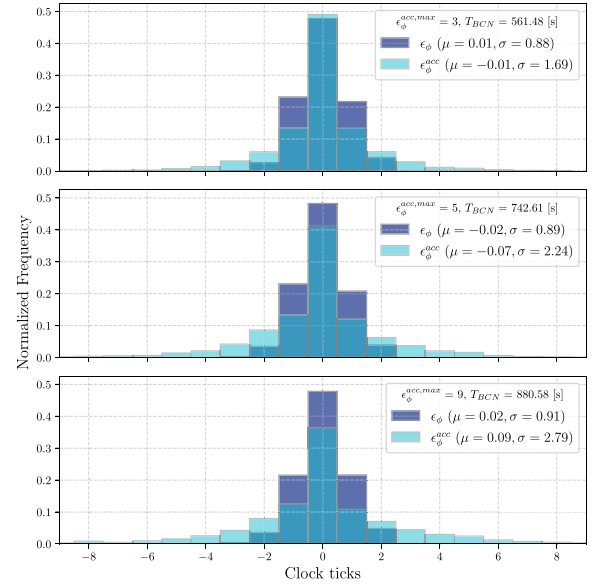


Fig. 13. Histograms of the errors distributions for different $\epsilon_z^{\text{acc}, \max}$ limits, while $\Delta T_{\max}$ is fixed at 0.3°C . By limiting temperature changes at each step, the step error remains constant. The dispersion of the accumulated error, instead, increases as its threshold, $\epsilon_z^{\text{acc}, \max}$, increases.

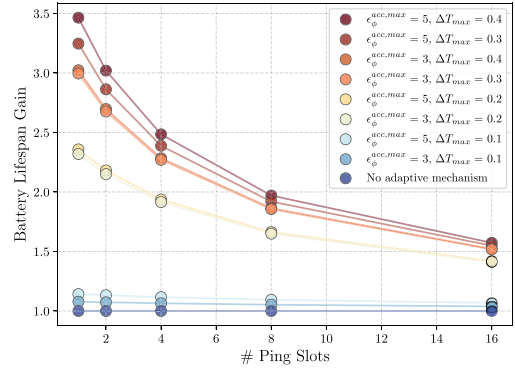


Fig. 14. Battery lifespan gains estimated using the model in (2) and the beacon period measurements obtained during Experiment 2.

contrast, with the proposed adaptive mechanism, the accumulated offset per step is nearly zero, significantly improving interbeacon synchronization accuracy between the End device and the gateway or time source. In fact, the mean error is reduced by more than three orders of magnitude. Furthermore, across all devices in this experiment, the average beacon period exceeded 12 min, leading to fewer received beacons and, consequently, lower power consumption. Notably, the low computational cost of the mechanism makes it suitable not only for LoRaWAN Class-B systems but also for other beacon-based synchronization systems.

1) *Device Calibration:* For applications that demand higher accuracy, precalibration of the device is required. This can be achieved by performing a series of drift measurements across the entire temperature range and applying a linear regression to determine the parameters in (19). By doing so, the uncertainties in γ' and T'_0 can be reduced, thus removing $\epsilon_{\hat{q}}$ from (27) (for further details, refer to [31]).

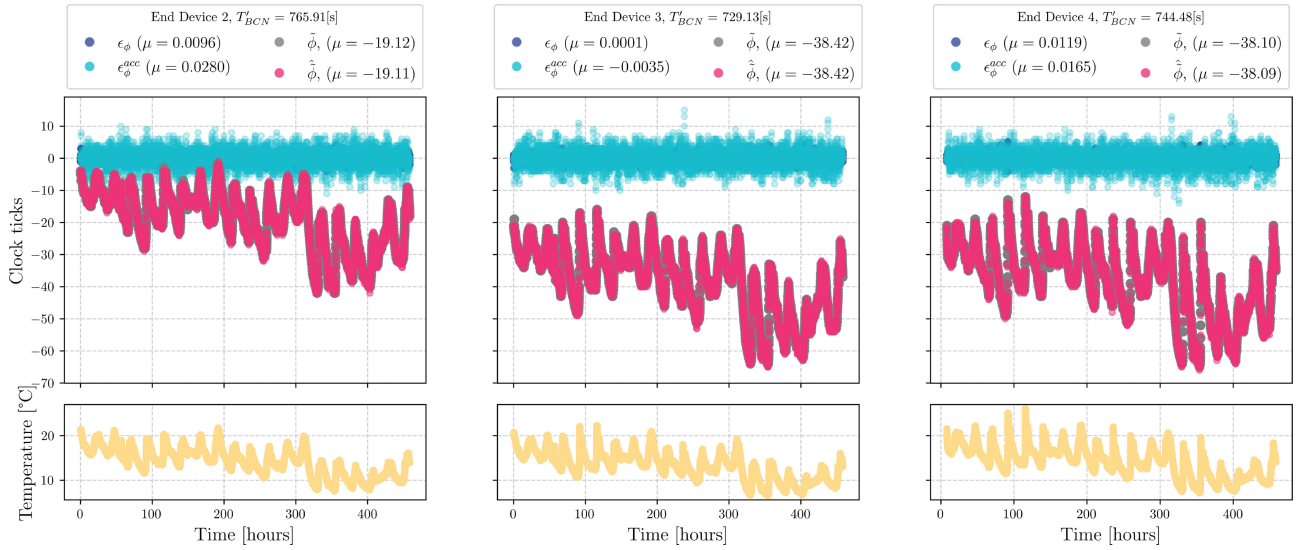


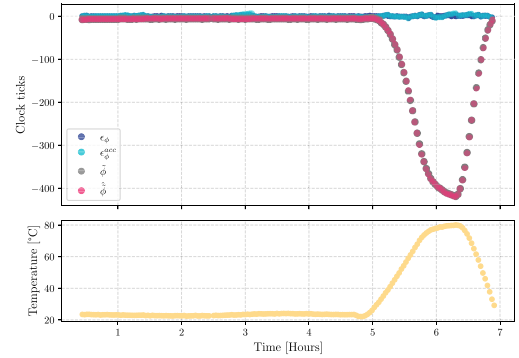
Fig. 15. Long-term deployment results for three different End devices over a test duration of more than 400 h. In all cases, the devices were configured with the following parameters: $\Delta T_{max} = 0.5^\circ\text{C}$, $\epsilon_{\phi}^{acc,max} = 4$, and $\delta T_{max} = 5^\circ\text{C}$. These parameters were maintained throughout the entire testing period to evaluate the performance and stability of the devices under sustained operation.

2) *Beacon Loss*: Although beacons are transmitted on dedicated channels, separate from regular data, and protected by guard intervals to prevent collisions, there remains a small chance that an End device may miss one. In such cases, the device continues predicting using previous data, while the synchronization error accumulates until the arrival of the following beacon. From Fig. 15, End device 2 shows an average accumulated error of $\epsilon_{\phi}^{acc} = 0.028$ ticks over 765.91 s, which is equivalent to an effective drift of 0.001 ppm. Thus, the error after p lost beacons grows linearly as $\epsilon_p = p \times T_{BCN} \times 0.001$ ppm.

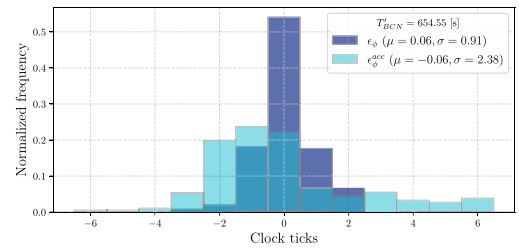
D. Experiment 4: Extreme Environment

Fig. 16 shows the results of an experiment simulating a stable environment that undergoes sudden temperature changes exceeding $2^\circ\text{C}/\text{min}$ for a duration of 2 h. To replicate such conditions, the End device was placed inside a climatic chamber. The results illustrate how significant clock drift can become under extreme conditions. In some 128 s intervals, the offset can exceed 420 ticks (approximately 12 ms) if no drift compensation is applied. In contrast, with the proposed adaptive mechanism enabled, the accumulated error remains bounded within 6 ticks, emphasizing the importance of actively compensating for clock drift.

The conditions during the first 6 h of the experiment resulted in an average beacon period of 989.0 s. However, as shown in the histogram, the abrupt temperature changes during the final 2 h reduced the overall effective beacon interval, T'_{BCN} , to approximately 11 min. This highlights both the system's ability to adapt to sudden environmental changes, by reducing the number of prediction steps, k , to maintain synchronization accuracy, and the tradeoff it incurs: more restrictive conditions limit the achievable energy savings compared to more stable environments.



(a)



(b)

Fig. 16. Experimental results from subjecting an End device to sudden and extreme temperature variations inside a climatic chamber. In the final 2 h of the experiment, the crystal exhibited a drift of up to 95 ppm, potentially causing synchronization errors exceeding 12 ms within just 128 s. In contrast, the proposed adaptive mechanism effectively bounds this error to a maximum of 6 ticks or, equivalently, $183\mu\text{s}$. (a) Time evolution of clock tick ground truth and predictions. (b) Histogram of the step and accumulated errors.

E. Comparative Analysis

The previous sections demonstrated the improvements offered by our synchronization mechanism over the LoRaWAN Class-B standard. We now take a broader

TABLE III
COMPARISON OF VARIOUS SYNCHRONIZATION METHODOLOGIES UNDER EQUIVALENT BATTERY LIFESPAN CONSTRAINTS

Mechanism	Max. Accumulated Synchronization Error (30 min period)	Network Overhead Introduced	Estimated Model Update Operation Count	Implementation Type	Pre-deployment Requirements	Long-Range Capabilities
This Work	610 μ s	Low	16	Sim. & Outdoor Testbed	None	Yes
Martinez et al. [31]	2 ms	Medium/High	80	Simulation	Pre-calibration	No
Elsts et al. [32]	.5 ms	Medium/High	30	Sim. & Outdoor Testbed	Pre-calibration	No
Masood et al. [33]	10 ms	Low	12000	Outdoor Testbed	None	No
Lübken et. al [34]	1.6 ms	Low	20	Sim. & Indoor Testbed	Crystal characterization	No

perspective, highlighting how the specific characteristics of our approach make it particularly well-suited for SHM applications, in comparison with existing works.

To ensure a fair comparison, we selected results from previous works where the average beacon period achieved is approximately 30 min, ensuring comparable energy savings across all methods. Under this equalized energy budget, we then focused on comparing synchronization accuracy and other key characteristics relevant to industrial applications.

Table III presents a comparative analysis between our method and several representative mechanisms from the literature. One of the most notable results is that our method achieves the lowest maximum accumulated synchronization error over a 30 min period. While other methods may show acceptable average errors, our focus is on worst-case performance to ensure robust system behavior under all conditions.

Another important aspect is network overhead. Methods, such as those in [31], [32] rely on periodic request–response exchanges initiated by the nodes, which scale poorly as network size increases. In contrast, our approach passively leverages LoRaWAN Class-B beacons, which are already periodically broadcasted by the gateway for synchronization. As these beacons are shared across all nodes, no additional uplink traffic is generated, preserving scalability. Moreover, LoRaWAN specifies dedicated channels for these control-type frames, reducing contention with user data and minimizing network congestion.

Predeployment requirements are another limiting factor for some techniques. For example, the approach in [34] achieves excellent accuracy and low overhead, but it requires exhaustive precharacterization of the oscillator for each device, limiting scalability and generalization to diverse hardware platforms. Our mechanism requires no such calibration or device-specific tuning.

Finally, it is important to note that all the aforementioned implementations depend on physical layers not designed for long-range communication, which limits their applicability to SHM scenarios involving large infrastructures or remote locations. In such cases, multihop extensions would be required, increasing network complexity and introducing additional synchronization errors. In contrast, our approach is natively compatible with the LoRa PHY, maintaining the long-range communication features essential for typical SHM deployments.

VIII. CONCLUSION

In this work, we have proposed an adaptive mechanism to enhance the energy efficiency of LoRaWAN Class-B networks. Class-B, a specific operational mode defined in the LoRaWAN standard, is particularly valuable when precise synchronization is critical. However, its energy-intensive nature makes it less suited for low-power devices. Our lightweight mechanism addresses this limitation by predicting clock drift, even under varying temperature conditions, to maintain device synchronization while extending the beacon period. By reducing the number of beacon receptions, the mechanism significantly lowers the energy demands of Class-B mode, making it feasible for use in battery-powered devices. To validate the system's effectiveness, we have implemented it on commercial IoT devices with a minimal memory footprint of 2.5 kB. Extensive experimental evaluations show that our dynamic technique can more than triple device battery life while reducing interbeacon synchronization error by over three orders of magnitude, on average. These results position our solution as a highly valuable tool for large-scale asset monitoring systems, where long-range communications, low-power consumption, and measurement synchronization are required.

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Luca Scalambri (Graduate Student Member, IEEE) received the Engineering degree in telecommunications from Balseiro Institute, National University of Cuyo, Mendoza, Argentina, in 2021. He is currently pursuing the Ph.D. degree with the Universitat Oberta de Catalunya, Barcelona, Spain, under a Marie Skłodowska-Curie Fellowship.

He works as an Innovation and Research Engineer with Worldsensing SL, Barcelona. His research interests include low-power wireless networks, the LoRa PHY layer, and embedded systems.

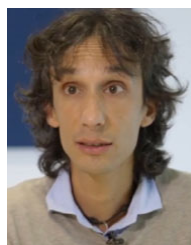


Andrea Zanella (Senior Member, IEEE) received the Laurea degree in computer engineering and the Ph.D. degree from the University of Padova, Padua, Italy, in 1998 and 2001, respectively.

He is a Full Professor with the Department of Information Engineering, University of Padova. During 2000, he was Visiting Scholar with the University of California at Los Angeles, Los Angeles, CA, USA, within Prof. Mario Gerla's research team. He has authored 220+ papers. He long-established research activity is in the field of

protocol design, optimization, and performance evaluation of wired and wireless networks.

Prof. Zanella has been serving in the editorial board of several top journals, such as IEEE INTERNET OF THINGS JOURNAL, IEEE COMMUNICATIONS SURVEYS AND TUTORIALS, and IEEE TRANSACTIONS ON COGNITIVE COMMUNICATIONS AND NETWORKING.



Xavier Vilajosana (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees in computer science from the Universitat Politècnica de Catalunya, Barcelona, Spain, in 2004 and 2006, respectively, and the Ph.D. degree in computer science from the Universitat Oberta de Catalunya (UOC), Barcelona, in 2009.

He has been a Researcher with Orange Labs, University of California, Berkeley, Berkeley, CA, USA, and HP Labs, Palo Alto, CA, USA. He is currently a Professor with UOC. His research interests include wireless communications and standardization.